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Structural Instabilities in Lanthanum Cuprate Superconductors

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Subtle but well characterized structural phase transformations occur at low temperatures in some lanthanum cuprate superconductors. These transformations, which involve collective tilting of the CuO_6 octahedra, have a strong influence on superconducting and normal state properties. Experiments suggest that for a given hole doping the following hierarchy exists for superconductivity in these materials: $T_C(HTT) > T_C(LTO) > T_C(LTT)$, where HTT represents the untilted structure and LTO and LTT are the two limiting low temperature tilted structures. An additional very strong suppression of T_C, suggestive of a competing electronic groundstate, occurs in the LTT phase for a hole to Cu ratio of 1/8. Neither of these effects are understood at present. This paper reviews both the evidence upon which these conclusions are based and some outstanding unanswered questions.

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1. INTRODUCTION

Cuprate superconductivity was first found by Bednorz and Müller in doped La_2CuO_4.[1] On account of its relative simplicity, the La_2CuO_4 structure is considered the archetype of the many other structures with CuO planes that exhibit high temperature superconductivity. There is, however, not only an undistorted La_2CuO_4 structure, but also several closely related distorted La_2CuO_4 structures, depending upon composition, temperature, and pressure. For reasons that are not yet clear, these subtle structural modifications cause substantial changes in the superconducting properties and have therefore attracted considerable interest.

In Sections 2 and 3 of this paper we review in some detail the structural relationships of these materials. The influence of these structural rearrangements on the normal state electronic transport properties and on the superconductivity of the doped metals are discussed briefly in Sections 4 and 5, respectively. The structural changes also influence the antiferromagnetic structure of the undoped insulators, and this is reviewed in Section 6. Finally in Section 7 we raise some outstanding questions that need to be answered to achieve a better understanding of the

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significance of structural instabilities for cuprate superconductivity.

2. STRUCTURAL RELATIONSHIPS IN $La_{2-x}Ba_xCuO_4$

In essence, the La_2CuO_4-structure is lamellar, consisting of CuO_6 octahedra arranged in a checkerboard pattern with in-plane oxygen atoms belonging simultaneously to two octahedral units, as shown in Figure 1. The remaining oxygen

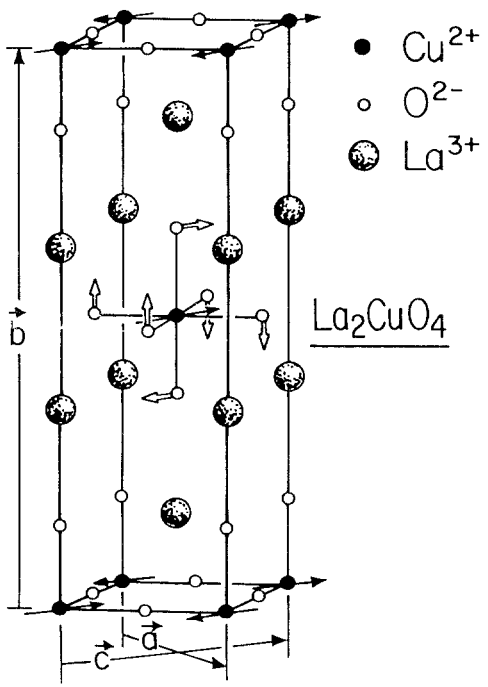


Fig. 1. The crystal structure of $La_2CuO_4$with the directions adopted by the magnetic moments of Cu^{2+} illustrated by arrows. The magnetic structure is that of the LTO phase.

atoms form rocksalt-like layers of LaO above and below the CuO_2 planes. The planes are flat and regular in the high temperature tetragonal (HTT) modification. At $T_O \simeq 530$ K La_2CuO_4 undergoes a second order phase transformation to a low temperature orthorhombic (LTO) form. The LTO form basically differs only by a slight cooperative tilting of the CuO_6 octahedra about one of the two [110] axes of the HTT form.[2] The size of the unit cell doubles, and new superlattice peaks appear with intensities – proportional to the square of the tilt angle – which increase smoothly with decreasing temperatures below T_O.

The atomic displacements involved in the HTT→LTO transformation are just those associated with a degenerate pair of Brillouin zone-boundary vibrational modes. An inelastic neutron scattering study of the phonons in La_2CuO_4 showed that the frequencies of these phonon modes were anomalously soft and decreased essentially to zero in the vicinity of the transformation temperature, as shown in Figure 2, a classical example of a soft mode phase transformation.[3] The LTO phase

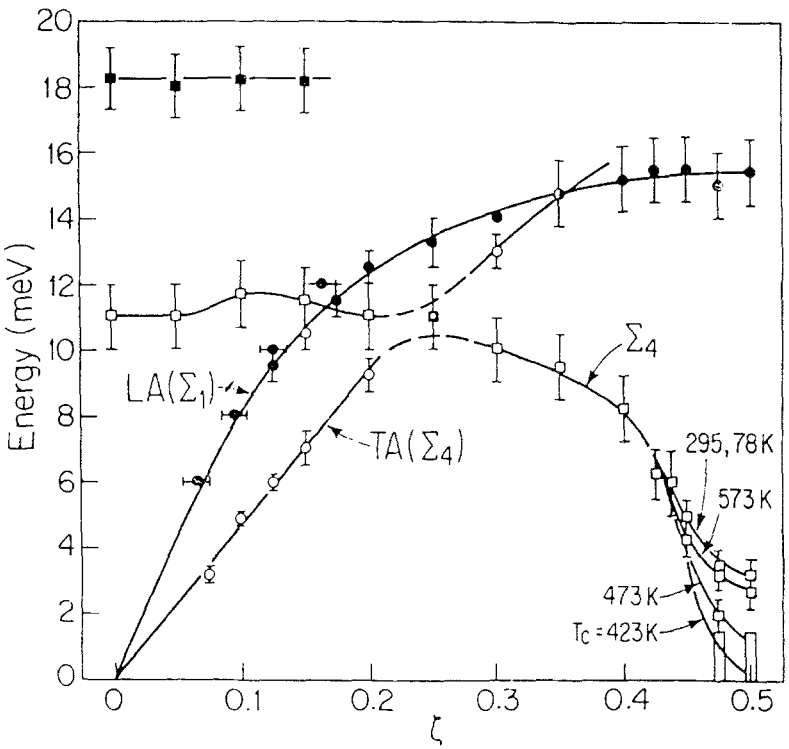


Fig. 2. Low energy phonon dispersion curves for La_2CuO_4 determined by inelastic neutron scattering measurements. Note the soft CuO_6 octahedra rotational mode at the Brillouin zone boundary X point.[3]

exists in two structural modifications (twin domains), resulting from the condensation of one or the other of the pair of degenerate phonon modes.

The first cuprate superconductors were made by doping La_2CuO_4 with divalent hole-donors.[1] Further structural complications were discovered by Axe et al. in Ba-doped La_2CuO_4, which over a suitable range of compositions transformed from the LTO structure to a low temperature tetragonal (LTT) structure, at temperatures $T_t \leq 70$ K[4] (See Figure 3). The LTT structure was predicted[4] and shown by neutron powder diffraction[5] to represent the equal condensation of both degenerate soft phonons – i.e. a coherent superposition of LTO twins – in which the tilt axes of the octahedra are rotated to lie along [100] axes of the HTT phase (Figure 4).

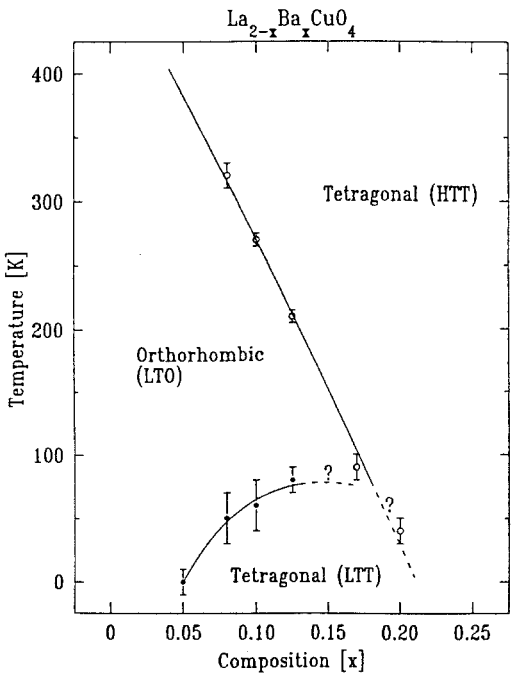


Fig. 3. The structural and superconducting phase diagram of $La_{2-x}Ba_xCuO_4$. The structural data are from x-ray powder diffraction measurements, T_C from magnetic susceptibility.[4]

The low temperature atomic coordinates were determined by Cox et al. by neutron diffraction[5] (See Table 1). The LTO→LTT transformation is discontinuous (first order), but the magnitudes of the octahedra tilts are little affected by the transformation – only the tilt direction changes.

Landau theory is helpful in understanding the tilting instabilities of materials with the La_2CuO_4 structure.[4,6] The Hamiltonian symmetry is that of the XY model, in which the order parameters (Q_1,Q_2) represent the tilts around the (110) and $(1\bar{1}0)$ axes of the HTT phase. The HTT→LTO transformation is caused by a vanishing of the lowest order (quadratic) order parameter couplings. The LTO→LTT transformation is caused by a subsequent temperature dependent interplay between quartic and eighth-order (and conceivably higher order) anisotropic terms in (Q_1,Q_2). Theory predicts that, in addition to HTT ($Q_1 = Q_2 = 0$), LTO ($Q_1 = 0$, $Q_2 \neq 0$) and LTT ($Q_1 = Q_2 \neq 0$) phases, under suitable conditions a fourth phase with ($Q_1 \neq Q_2 \neq 0$) would exist and interpolate smoothly between the LTO and LTT phases. This phase is sometimes called LTO2 and sometimes low temperature *less* orthorhombic (LTLO), since the orthorhombic distortion is proportional to $(Q_1^2 - Q_2^2)$.

Time dependent Landau theory has also been used to consider the nature of the soft mode fluctuations in the low temperature phases, i.e. below T_O and

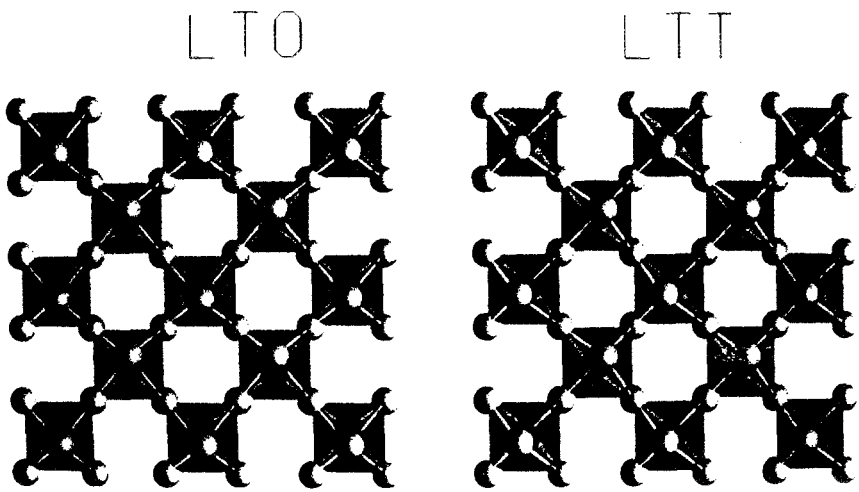


Fig. 4. A visual comparison of the LTO and LTT tilt structures of $La_{1.875}Ba_{0.125}CuO_4$. The perspective is looking along the c (long) axis onto the Cu-O plane. The tilt angles have been doubled to accentuate the sense of the tilts.

particularly in the vicinity of T_t.[6] The expected behavior is shown schematically in Figure 5. In the LTO phase the original HTT phonon degeneracy is split. The upper branch represents fluctuations in the *amplitude* of the tilting about the [110]-axes, and the lower branch is associated with fluctuations in the *direction* of the tilting away from the [110]-axes. (In the language of the XY-model, these represent longitudinal and transverse fluctuations of the order parameter, respectively.) The latter mode, whose stiffness reflects the relatively weak anisotropy energy, remains soft and near T_t this mode is expected to again become unstable. The detailed behavior depends upon whether the LTO→LTT transformation is direct or proceeds through an intermediate LTO2 phase. The upper mode, which folds back to the Brillouin zone center, has been observed in Raman scattering studies in La_2NiO_4.[7] The lower mode has been observed in lightly doped La_2CuO_4, but the interesting behavior near T_t has not been observed for lack of large single crystals of heavily doped material necessary for inelastic neutron scattering studies.[8]

3. THE STRUCTURE OF RARE-EARTH SUBSTITUTED La_2CuO_4

A more comprehensive experimental elucidation of the structural relationships was made difficult in $La_{2-x}Ba_xCuO_4$ by two facts. First, particularly for $x \geq 0.15$, the tilts are very small because of the convergence of the various structures (HTT, LTO, LTT). Second, variable but substantial amounts of LTO phase persist below T_t and coexist metastably with the LTT phase. This accounts for the uncertainties indicated by the dotted lines in Figure 3. These difficulties were largely circumvented with the discovery by Crawford and coworkers[9] that similar low temperature phase transitions can also be observed in insulating $La_{2-y}RE_yCuO_4$ (where RE stands for any of the light rare earth elements) and their hole-doped derivatives. Substitution

		$La_{1.9}Ba_{0.1}CuO_4$		$La_{1.475}Nd_{0.4}Sr_{0.125}CuO_4$	
		T=15 K	T=91 K	T=10 K	T=100 K
Cu:	B(Å^2)	0.0(1)	-0.1(1)	0.20(3)	0.16(2)
La:	x	0.0042(2)	0	0.0065(3)	0
	y	0.0042	0.0056(4)	0.0065(3)	0.0076(3)
	z	0.3612(1)	0.3609(1)	0.3611(1)	0.36091(8)
	B(Å^2)	0.0(1)	0.1(1)	0.05(3)	0.11(2)
01:	z	0.0081(4)	0.0054(2)	0.0106(4)	0.0075(2)
	B(Å^2)	-0.1(3)	0.3(1)	0.41(3)	0.40(3)
02:	x	-0.0213(3)	0	-0.0246(4)	0
	y	-0.0213	-0.0279(5)	-0.0246(4)	-0.0341(3)
	z	0.1824(2)	0.1825(2)	0.1834(2)	0.1829(1)
	B(Å^2)	0.6(1)	0.7(1)	0.60(4)	0.70(3)
	a(Å)	5.3548(1)	5.3436(1)	5.34083(8)	5.31528(7)
	b(Å)	5.3643(1)	5.3773(1)	5.34083(8)	5.36900(7)
	c(Å)	13.2356(3)	13.2386(2)	13.1063(3)	13.1121(2)
CuO_6 tilt angle:		3.8°	3.6°	4.4°	4.4°

Table 1
Powder neutron diffraction data for $La_{1.9}Ba_{0.1}CuO_4$[5] and $La_{1.475}Nd_{0.4}Sr_{0.125}CuO_4$[49]

of smaller rare earths for lanthanum stabilizes the tilted structures relative to the untilted one, and promotes larger tilts at low temperatures. Metastable coexistence at low temperatures is also largely eliminated.

The properties of many RE-doped and RE-(Sr or Ba) co-doped materials have been studied and reported.[9–11] A representative phase diagram is shown in Figure 6. Note the appearance of the predicted LTO2 phase which interpolates between the LTO and LTT phases. As anticipated by Landau theory,[4] LTO2 vanishes at a bicritical point where all three tilted phases are in equilibrium, corresponding to the vanishing of both quartic and eighth order anisotropy.[6] In Figure 7 we show synchrotron x-ray powder diffraction scans of the $(110)_{HTT}$ reflection for $La_{1.49}Nd_{0.4}Sr_{0.11}CuO_4$, which has LTO and LTT phases at the highest and lowest temperatures and coexisting LTO and LTO2 phases at 57 K.[9,10] The low temperature phase stability relations are also influenced by small departures from oxygen stoichiometry. The only insulating, oxygen stoichiometric RE-doped materials studied in detail are $La_{1-y}Nd_yCuO_4$, which pass through LTO to LTO2 to LTT low temperature configurations, with increasing Nd doping.[12]

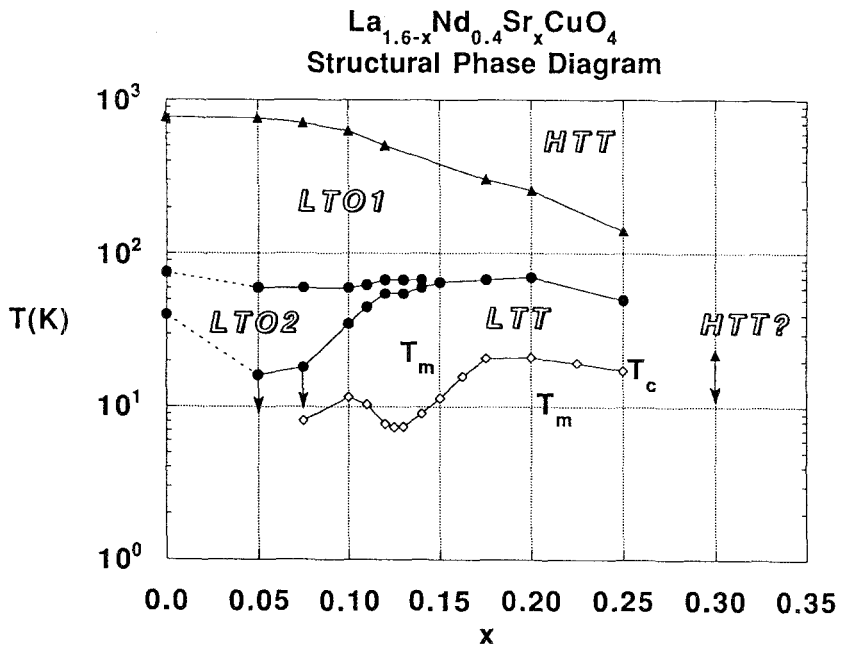


Fig. 5. The structural and superconducting phase diagram of $La_{1.6-x}Nd_{0.4}Sr_xCuO_4$. The structural data are the result of high resolution synchrotron x-ray powder diffraction measurements. The T_m are magnetic ordering temperatures estimated from muon spin resonance measurements.[10]

4. STRUCTURE AND NORMAL STATE PROPERTIES

Being phenomenological, the Landau theory provides no microscopic understanding of the low temperature transformations in La_2CuO_4-based compounds. But since such transformations are also known to exist in isostructural insulating compounds with very different compositions (e.g. $La_{2-y}RE_yCuO_4$, other oxides such as La_2NiO_4[13] and La_2CoO_4,[14] as well as chlorides of the form A_2MCl_4, where $A=CH_3NH_4^+$, etc. and $M=Mn^{2+}$, etc.,[15] it is probably unnecessary to invoke specific features of the band structure to provide a driving force, as suggested by some authors.[42] Inherent ionic size differences between the alternating rock-salt and perovskite-like layers may be sufficient to provide the major part of the stabilization.

On the other hand, the subtle low temperature transformations do have an important effect on the normal state transport properties of these materials, and for this reason there have been numerous studies and speculations concerning their interrelation. Perhaps the first study was that of Sera et al.,[17] in which several normal state properties, including resistivity, magnetic susceptibility and thermopower were measured for $La_{2-x}Ba_xCuO_4$ ceramic samples for temperatures spanning the LTO→LTT structural transformation. All of the transport properties showed anomalies at or near the structural transformation temperature when x=0.12. In-

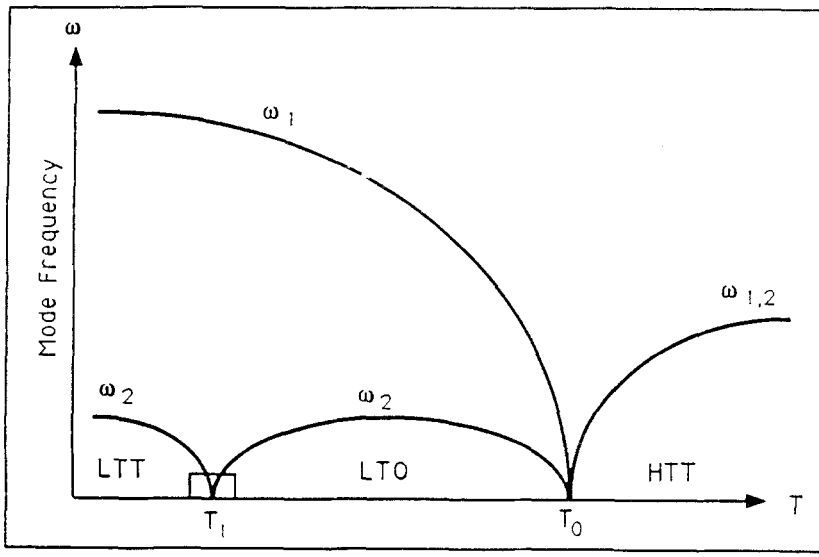


Fig. 6. A schematic diagram of the CuO_6 octahedra tilt mode energies as a function of temperature through the various structural phase transformations. Above T_O there are two degenerate modes, representing tilts about orthogonal axes. Below T_O the modes are re-diagonalized, representing fluctuations in tilt amplitude (ω_1) and tilt direction (ω_2), respectively. ω_2, reflecting the weak anisotropy of the system, softens again near T_t.[6]

terestingly, the increase of the resistivity and the decrease of the magnetic susceptibility at the transformation temperature could not be consistently explained (even qualitatively) simply by a change in the density of states at the Fermi surface due to the structure change.

Specific heat measurements[11,18] also show sizable anomalies at the LTO→LTT and LTO→LTO2 transformations in $La_{2-x-y}Nd_ySr_xCuO_4$, whereas there is a much smaller specific heat anomaly[19] at the LTO→LTT transition in $La_{1.875}Ba_{0.125}CuO_4$. This observation is consistent with expectations based upon Landau theory, which suggest that the magnitude of the specific heat anomaly should vary as the fourth power of the tilt angle of the octahedra, which is generally larger in the more highly doped Nd^{3+} substituted $La_{2-x-y}Nd_ySr_xCuO_4$ samples than in $La_{1.875}Ba_{0.125}CuO_4$ (Table 1).

Although there have been many other studies of Ba doped ceramic materials, to date the only study of the influence of the LTO→LTT transition on the *anisotropic* transport properties in these systems is that on single crystals of $La_{1.6-x}Nd_{0.4}Sr_xCuO_4$ by Nakamura and Uchida.[20] Anomalous discontinuities at the structural phase transformation are observed in the ab-plane and in the c axis directions. Furthermore, the anomalies were far more pronounced for the x=0.125 sample than for the x=0.2 sample, and the latter was observed to be a bulk superconductor, whereas the x=0.125 sample was virtually nonsuperconducting. This

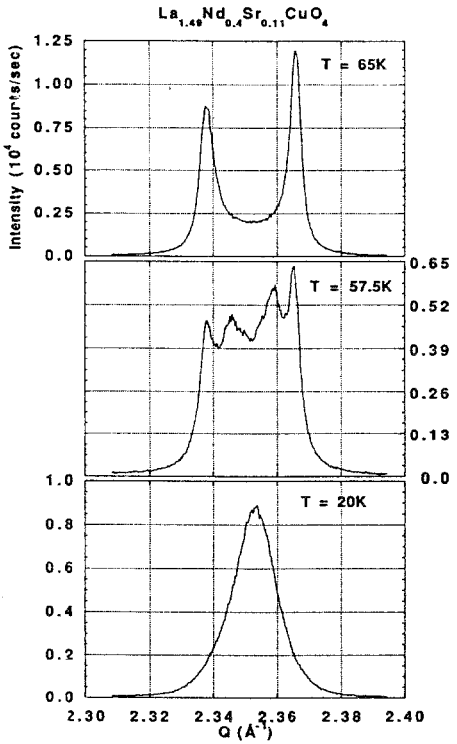


Fig. 7. Synchrotron x-ray powder diffraction scans of the $(110)_{HTT}$ reflection of $La_{1.49}Nd_{0.4}Sr_{0.11}CuO_4$, showing orthorhombic splitting characteristic of LTO, LTO+LTO2 and (no splitting) LTT structures, respectively, as the temperature is decreased.[9,10]

final point will be discussed further in the following section.

5. STRUCTURE AND SUPERCONDUCTIVITY

Unlike in $La_{2-x}Sr_xCuO_4$, bulk superconductivity in $La_{2-x}Ba_xCuO_4$ is dramatically suppressed in the region of composition around x=0.125, as first discovered by Moodenbaugh et al.[4,21] Moreover, several observers[4,22] noted an apparent inverse correlation between the fraction of transformed LTT phase and the amount of superconductivity exhibited by these materials. These initial observations of the suppression of T_C and the appearance of the LTT phase led to the natural suggestion that the LTT phase was unfavorable for, and possibly destroyed, superconductivity. This correlation, however, was complicated by the difficulty, even with high resolution synchrotron x-ray diffraction, of determining the precise phase composition of all the samples, particularly at doping levels $x \geq 0.15$, as discussed in Section 3.

It was partly to rectify this difficulty that experiments on samples which were substituted with various rare earths on the La site were initiated by Crawford et al.[9] Even more important, by adjusting isovalent (RE) and heterovalent (Sr, Ba) dopants it was possible to vary the doping level and the tilt structure *independently*, and thus the influence of each of these factors on the superconducting phase transformation could be evaluated.

These and other similar experiments[20,23–25] clearly demonstrated that 1) the onset of superconductivity is suppressed by the LTO→LTT transformation and 2) there is, in addition, a very strong influence of the precise hole concentration. In particular, T_C is greatly suppressed for hole concentrations near x=1/8. Although samples which are heavily doped with rare earths are not superconducting (an effect possibly related to the large tilt angles in these materials[10,11]), it is now clear that both the LTT and LTO phases can, under appropriate circumstances, support bulk superconductivity. From a consideration of the results of high pressure measurements of T_C[26] and tilt structure,[27] it has been suggested that the following hierarchy exists for superconductivity in these materials *at the same hole concentration*: $T_C(HTT)>T_C(LTO)>T_C(LTO2)>T_C(LTT)$.[10] A number of studies have arrived at conclusions similar to these,[23–25] although there is still some controversy over these issues.[28]

Although, as discussed earlier, it seems unlikely that the LTT phase is substantially stabilized electronically, the suppression of superconductivity and the normal state transport anomalies near the doping level x=1/8 are very striking. They must be the consequence of some as yet not understood electronic effects controlled by the hole concentration and strongly enhanced in the LTT phase. A number of studies support the identification of x=1/8 as a special hole concentration. The chemical substitution studies of Maeno et al.[29] offered early and very convincing evidence. Some additional important early indications came from other chemical substitution studies,[30] oxygen isotope effect studies,[31] and oxygen stoichiometry studies.[23] Although an isotope effect conventionally signals the participation of phonons in the superconductive pairing, the large enhanced isotope effect at x=1/8 in $La_{2-x}Sr_xCuO_4$ and $La_{2-x}Ba_xCuO_4$ is not easily explained by a conventional phonon pairing mechanism, and a definitive theoretical explanation of this phenomenon is still lacking. Nevertheless, it seems to us that the case for a unique (probably electronically stabilized) ground state at x=1/8 is now well established.

A number of high pressure studies have been reported for doped La_2CuO_4, and these results raise several important questions. While not in precise quantitative agreement, the experiments support the qualitative conclusion that the HTT→LTO (or LTO2) transition and the LTO→LTT transition temperatures decrease with increasing pressure, and that the superconducting T_C increases when the LTT phase is suppressed by pressure.[26,27] However, even at 20 kbar some suppression of T_C is still evident[26] when x=1/8. This is surprising, since thermal expansion measurements suggest that the materials are in the HTT phase at this pressure.[26] This result emphasizes the need for a direct structure determination at high pressures (20 kbar). It is very important to resolve this issue since, for example, theoretical explanations which rely upon spin orbit coupling to suppress T_C will presumably not work in the HTT phase.[32]

6. STRUCTURE AND MAGNETIC PROPERTIES

The cuprates in their insulating forms are well described as two dimensional Heisenberg antiferromagnets.[33] Because the Cu^{2+} ion is spin 1/2, the leading order magnetic anisotropy is a result of antisymmetric superexchange, generally referred as the Dzyaloshinsky-Moriya (D-M) interaction.[34,35] This anisotropic interaction is present only when there is sufficiently low symmetry in the structure, which is provided in these materials by the tilting of the octahedra. The D-M interaction favors the alignment of the spins in a plane defined by the c-axis and the direction of the tilt of the apical oxygens.[36]

The role of D-M interaction in the metamagnetism and spin-flop transitions in undoped La_2CuO_4has been described by several authors.[36,37] More recently, magnetic neutron diffraction measurements on samples of $La_{2-x}Nd_xCuO_4$ have shown that the structural transformation leading from LTO to LTO2 or LTT causes a Cu^{2+} spin reorientation in the basal plane, which seems to coincide with the structural transformation.[12,38,39] At lower temperature there is a distinct second magnetic transition in the LTO2 structure involving interlayer magnetic order (and apparently unrelated to structural rearrangements). The magnetic moments in every other layer rotate by 180° in the basal plane leading from a metamagnetic state to a weakly ferromagnetic one.[12] The LTT phases are also weak ferromagnets.[38,39] These spin reorientations reflect a change in the sign of the effective magnetic interlayer coupling at lower temperatures in the LTO2 and LTT structures.

The magnetic interlayer interactions, and in particular the effect of crystal symmetry upon these interactions, are far less well understood than the intralayer (or single layer) magnetic coupling. In fact, the dominant source of interlayer coupling (antisymmetric exchange, dipolar coupling?) in these materials is little discussed, and may change with tilt structures.

7. REMAINING QUESTIONS

Despite the considerable experimental and theoretical effort which has been spent trying to unravel the effects of structural distortions on superconductivity in these materials, a number of unanswered questions remain.

7.1. Why is x=1/8 special?

Perhaps the most important question involves the origin of the unusual T_C suppression at a doping level of 1/8 hole per Cu^{2+} ion. Although we believe that the accumulated evidence leaves little doubt that a unique state occurs rather precisely at a hole concentration of 1/8, there is no consensus as to why 1/8 is special. The appearance of a rational fraction is reminiscent of a commensurability effect due to formation of a charge or spin ordered state, and several authors have suggested such a scenario.[29,31] Nevertheless, there have been no direct observations of such ordering, although some evidence of charge ordering has recently been reported in Sr-doped La_2NiO_4.[40] The possibility that the Fermi surface intersects a van Hove singularity at x=1/8 has also been suggested,[41–43] but there is no corroborating evidence in the form of enhanced normal state Pauli susceptibility or specific heat.

7.2. Why is T_C so sensitive to tilts?

There have been several suggestions to account for the suppression of T_C due to the CuO_6 tilting transformations, ranging from a decrease in the density of hole states at the Fermi surface[44] to a stabilization of a commensurate antiferromagnetic metallic state.[32]

The initial report of a large decrease in the density of states derived from band structure calculations[44] was based upon tilts which were roughly two times larger than the values determined from powder neutron diffraction (Table 1). More recent calculations,[45] using the correct tilt angles,[5] show little change in the density of states as a result of the LTO to LTT transformation and thus cast doubt on this explanation of the T_C suppression.[46] These conventional orbital overlap effects scale as the *square* of the tilt angle, which typically varies monotonically with the average size of the unit cell as it changes with atomic dopant concentration. They do not seem capable of explaining the anomalous electronic doping dependence near x=1/8. This strongly implies that additional factors, outside those considered by one electron band theory, are controlling the suppression of T_C.

Another explanation for the T_C suppression relies upon the spin-orbit effect, and a resultant coupling of the Cu^{2+} spins to the mobile holes through D-M-like terms linear in the tilt fluctuations, to cause pair breaking,[47] and to stabilize a non-superconducting antiferromagnetic ground state.[32] The theory predicts an upper critical tilt angle, beyond which superconductivity is suppressed. Büchner[11] has discussed experimental evidence in support of this idea. Other workers have concluded that these effects are too small to be important.[48] Contrary to observation, the theory makes no distinction between LTO- and LTT-like lilts, but the clear and direct involvement of the tilts in this mechanism is, in our opinion, a strong motivation for continued interest in this model. Additional support for a competing magnetic state comes from muon spin resonance and NMR experiments which show that some type of weak magnetic order exists in $La_{1.875}Ba_{0.125}CuO_4$ below 32 K.[49]

7.3. Is any of this relevant to high T_C in general?

This question arises in the context of the large amount of generally inconclusive work on the connection of structural phase transformations to high temperature superconductivity. Very few of the known high T_C materials exhibit well defined tilting phase transformations, and none of the highest T_C materials have been reported to show such a pronounced dependence of superconductivity on structural modifications. Nevertheless, we believe that there is much to be learned from studying materials which exhibit such subtle interplay of structure and function. Among all of the high T_C materials, those based upon La_2CuO_4 are arguably the simplest in terms of control of the hole concentration, an extremely important parameter in the cuprate superconductors. Furthermore, their crystal structures are relatively simple (single Cu-O layers, with minimal site disorder or unintentional oxygen non-stoichiometry). It seems to us unlikely we can really claim to understand high temperature superconductivity until we understand how such small changes in crystal structure and symmetry can profoundly influence the superconducting state. Perhaps, after all, nature has been kind in presenting us with such a well-posed problem lying so near to the heart of the high T_C enigma.

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